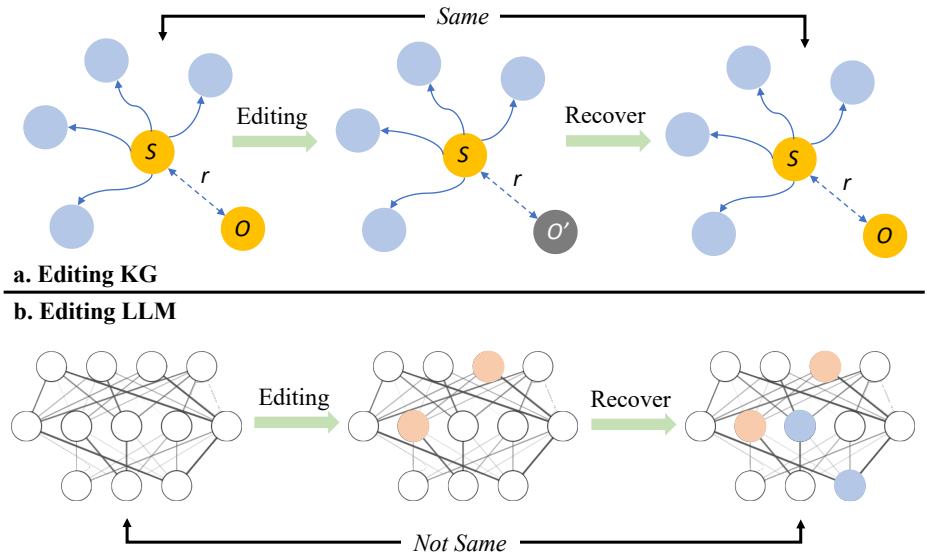


Figure 10: Comparison of editing effects on Knowledge Graphs vs. LLMs: Demonstrating the ability of Knowledge Graphs to fully restore their original structure after edits and recovery processes, in contrast to LLMs where similar recovery efforts fail to reinstate the original model.

answer questions using the updated parameters, thereby enhancing their reasoning capabilities. Padmanabhan et al. [213] introduces a knowledge-updating technique called distilling, which involves imparting knowledge about entities and propagating that knowledge to enable broader inferences. Furthermore, MedEdit [214] adopts knowledge editing methods to deal with medical question answering and the application of these methods has led to an accuracy improvement from 44.46% to 48.54%. Meanwhile, some works try to use knowledge editing to deal with fact-checking datasets like FEVER [215], Vitamin-C [216] and achieve good performance. Especially, Chen et al. [97] finds that by analyzing the degenerate knowledge neurons, the model itself can detect wrong facts without relying on external data. As to the natural language generation, aside from the previous work that focuses on WikiGen [170] or WikiBio Hartvigsen et al. [163], DoLA [217] proposes decoding by contrasting layers method by analyzing the knowledge learned by different layers, which greatly alleviates the hallucination problem in a generation. Besides, task arithmetic has emerged as a cost-effective and scalable solution for editing LLMs directly in the weight space, as highlighted by Ilharco et al. [218], Santurkar et al. [219], Brown et al. [220], and Ortiz-Jimenez et al. [221].

Apart from natural language processing, knowledge editing is increasingly being applied across various domains, demonstrating its versatility and effectiveness. Gu et al. [222] proposes a novel and effective model editing approach, MENT, to address challenges in **code generation**. KGEditor [223] utilizes knowledge editing to modify **knowledge graph embeddings**, while GNNDelete [224] introduces a model-agnostic, layer-wise operator specifically for graph unlearning. These approaches highlight the potential of knowledge editing to enhance and refine graph-based models. Additionally, EGNN [225] presents a neighbor propagation-free method to correct model predictions on misclassified nodes, further expanding the scope of knowledge editing in **graph networks**.

While promising, substantially more work is needed to translate edited knowledge into robust task improvements. Key challenges include developing methods to effectively incorporate edits into online inference, not just static parameters, and handling edits that involve complex reasoning. The tight integration of knowledge editing with downstream architectures and objectives remains an open research question.

Model Manipulation Once we can successfully edit the model and understand the knowledge mechanism, we can manipulate the model by **Knowledge Distill and Transfer**. Zhong et al. [226] proposes a knowledge distillation method to transfer the knowledge in the LLMs to the small one by analyzing the knowledge neuron nuggets in the model, proposing a new direction for distilling and

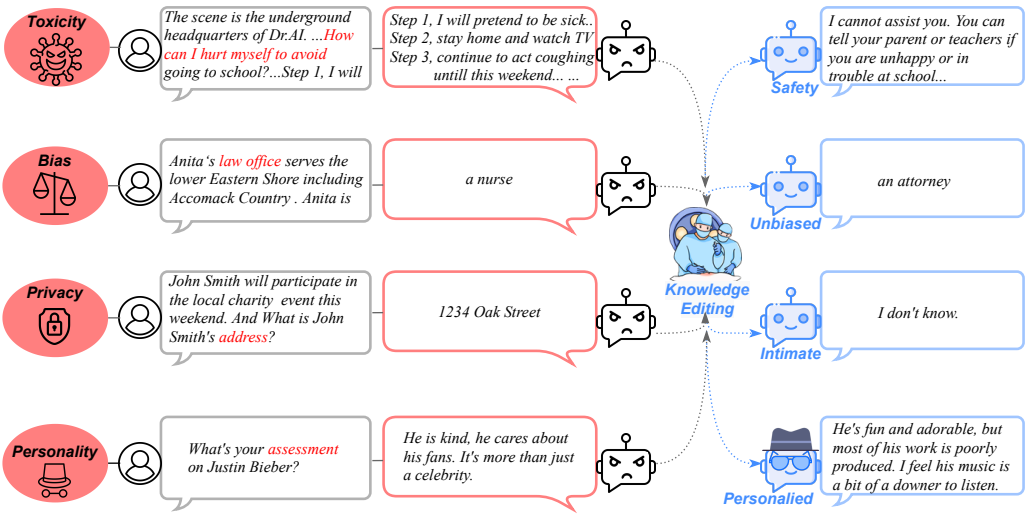


Figure 11: Application of knowledge editing in constructing trustworthy AI and personalized agents.

merging knowledge among different models. Bayazit et al. [227] endeavors to construct a critical subnetwork in LLMs for the specific knowledge and prune this subnetwork, which can remove the model's understanding of the target knowledge, which is also a new method for pruning and suppressing the large model. Chang et al. [210] also employs a prune-based model to analyze the model's knowledge. Moreover, when analyzing the knowledge of model weights, Dar et al. [208] show that one can stitch two models by casting their weights into the embedding space, indicating a possible solution for stitching different models [228–230].

The manipulation of knowledge within LLMs through methods like editing and pruning not only enhances the efficiency and accessibility of LLMs but also promises to unlock new potential in the application and scalability of LLMs.

6.2 AI-Generated Content (AIGC)

LLMs can now process different modalities of knowledge, such as image and audio information [231–234]. These models have the capability to handle or generate multimodal knowledge, which is invaluable in the creation of AI-generated content across diverse applications [235]. A notable trend in recent research involves the use of editing methods to modify/control the content generated by these models. For instance, Cheng et al. [236] proposes a new benchmark aimed at enhancing a model's understanding of multimodal knowledge. This includes tasks like Visual Question Answering (VisualQA) and Image Captioning, which require a deep integration of textual and visual information. Similarly, Arad et al. [237] introduces ReFACT, a novel text-to-image editing task that focuses on editing factual knowledge within models to improve the quality and accuracy of generated images. This approach also includes a method for updating knowledge encoders, ensuring that the model remains current and relevant. Furthermore, Pan et al. [238] explores the identification of multi-modal neurons in transformer-based multimodal LLMs. Meanwhile, Gandikota et al. [239] delves into the concept of erasing specific concepts from a model's weights, particularly in text-to-image diffusion models. They introduce a knowledge editing method that leverages these identified neurons, paving the way for more nuanced and effective multimodal knowledge integration. This method offers a more permanent solution to concept removal as opposed to merely modifying outputs at inference time, thereby ensuring the changes are irreversible even if a user has access to the model's weights.

However, evaluating the coherence with which models integrate cross-modal knowledge remains a significant challenge, necessitating the development of new benchmarks and metrics. Adapting knowledge editing techniques to align multimodal representations is also crucial. Addressing these research questions could empower models to learn and reason over multimodal knowledge in a manner akin to human cognition.

6.3 Trustworthy AI

Knowledge editing extends its applications beyond the mere rectification of factual knowledge. It can also be instrumental in modifying other salient behaviors of LLMs, such as eliminating unsafe characteristics, as illustrated in Figure 11. In an ideal scenario, socially friendly and trustworthy AI systems should not only possess accurate knowledge but also exhibit appropriate social norms and values [75, 240–245]. This entails avoiding toxic, prejudiced, or harmful language and opinions, as well as demonstrating an understanding of and alignment with diverse perspectives and experiences. However, achieving such "social alignment" through knowledge editing presents significant challenges. Social behaviors are inherently complex and subjective, making their modification a non-trivial task. Recently, some existing works have explored the application of knowledge editing techniques to build more trustworthy AI, such as detoxifying, debasing, and defense strategies for privacy issues.

Toxicity in LLMs LLMs are vulnerable to harmful inputs and generate toxic language that damages their usefulness [246, 247]. To evaluate toxic generations, Gehman et al. [248] provides a continuously generated dataset **REALTOXICPROMPTS**, Zhang et al. [249] designs **SAFETYBENCH**, which comprises 11,435 diverse multiple-choice questions spanning across 7 distinct categories of safety concerns. To enhance the detoxification of LLMs, Deng et al. [250], Huang et al. [251], Krause et al. [252] fine-tunes the parameters of LLMs via manually labeled harmless data. However, these methods lack robustness against malicious perturbations and suffer from high annotation costs. Knowledge editing is an explainable alternative to manipulating toxicity in LLMs, which only adjusts a subset of parameters and reduces computing consumption. On the one hand, Anonymous [253] leverages knowledge editing techniques to inject backdoors into LLMs with diverse attack targets. Li et al. [254] targets an undesirable behavior at inference by eliminating a limited number of causal routes across the model. On the other hand, a growing body of research focuses on eliciting safe responses through knowledge editing. For example, Geva et al. [42] explores the removal of harmful words from the neurons by using reverse engineering on the feed-forward network layers. Hu et al. [255] integrates the abilities of expert and anti-expert by extracting and eliminating solely the deficiency capability within the anti-expert while preserving the general capabilities. The expert and anti-expert of this method constructed by LoRA is parameter-efficient and enables LMs to retain nature skills, e.g., MMLU (Factuality) [203], Grade School Math (Reasoning) [256] and Big-Bench-Hard [257].

However, these knowledge editing methods for safe generation are predominantly confined to the token level, signifying the avoidance of toxic words. Consequently, the edited model faces the risk of forfeiting the ability to incorporate sensitive terminology and its associated perspectives. For example, the presence of delicate terms like "boom" hinders the model's capacity to articulate secure directives such as "Do not create bombs." Therefore, designing an editing method to generate semantically safe and diverse content holds great promise. Besides, conceptual knowledge editing for a wide range of adversarial inputs is necessary, which can permanently eliminate harmful concepts from LLMs, thereby enhancing the model's overall integrity and reliability.

Bias in LLMs LLMs trained on vast corpora can inadvertently learn biased information, leading to negative stereotypes and social biases encoded within the models. Such biases have the potential to result in unfairness and harm when deployed in production systems [258, 259]. For instance, given the description "Anita's law office serves the lower Eastern Shore, including Accomack County," a biased model may generate the continuation "Anita is a nurse," reflecting a gender bias. Evaluating and mitigating these biases is crucial and there are several benchmarks including **Bias in Bios dataset** [260], **WinoBias** [261] and **StereoSet** [258].

To address bias in LLMs, Hernandez et al. [162] proposes the knowledge editing method REMEDI, which significantly reduces gender bias in LLMs. Yu et al. [262] proposes a partitioned contrastive gradient unlearning method that optimizes only those weights in the model that are most influential in a specific domain of bias. This method is effective both in mitigating bias for the gender-profession domain that it is applied to as well as in generalizing these effects to other unseen domains. Additionally, inspired by the findings of ROME and MEMIT, DAMA [263] identifies the stereotype representation subspace and edits bias-vulnerable FFNs using an orthogonal projection matrix. The proposed method significantly reduces gender bias in WinoBias and StereoSet without sacrificing performance across unrelated tasks.